

## SCIKIT - LEARN

### 1. Classification

Identifying which category an object belongs to.

**Applications:** Spam detection, image recognition.

**Algorithms:** Gradient boosting, nearest neighbors, random forest, logistic regression.

### 2. Regression

Predicting a continuous-valued attribute associated with an object.

**Applications:** Drug response, stock prices.

**Algorithms:** Gradient boosting, nearest neighbors, random forest, ridge.

### 3. Clustering

Automatic grouping of similar objects into sets.

**Applications:** Customer segmentation, grouping experiment outcomes.

**Algorithms:** k-Means, HDBSCAN, hierarchical clustering.

### 4. Dimensionality reduction

Reducing the number of random variables to consider.

**Applications:** Visualization, increased efficiency.

**Algorithms:** PCA, feature selection, non-negative matrix factorization.

### 5. Model selection

Comparing, validating and choosing parameters and models.

**Applications:** Improved accuracy via parameter tuning.

**Algorithms:** Grid search, cross validation, metrics.

### 6. Preprocessing

Feature extraction and normalization.

**Applications:** Transforming input data such as text for use with machine learning algorithms.

**Algorithms:** Preprocessing, feature extraction.

## EVALUATION METRICS

For classification tasks on evaluation metrics include accuracy, precision, recall, F1-score and area under the ROC curve (AUC-ROC).

- **Accuracy:** Measures the proportion of correct predictions out of total predictions.
- **Precision:** Focuses on positive predictions, quantifying how many selected items are relevant.
- **Recall:** Also known as sensitivity, recall evaluates the model's ability to find all the relevant instances.
- **F1-score:** The harmonic mean of precision and recall, providing a balance between these two metrics.
- **AUC-ROC:** A metric for assessing the performance of a classification model where the output is a probability. It visually represents the tradeoff between the true positive rate (TPR) and false positive rate (FPR).

For regression tasks, common evaluation metrics in scikit-learn include mean absolute error (MAE), root mean squared error (RMSE),  $R^2$  score, and mean squared error (MSE).

- **MAE:** Measures the average magnitude of errors without considering their direction.
- **RMSE:** The square root of the mean of squared errors, giving more weight to larger errors.
- **$R^2$  score:** Also known as coefficient of determination, this score represents the proportion of the variance in the dependent variable that is predictable from the independent variables.
- **MSE:** Calculates the average squared difference between the predicted and actual values, offering a measure of how close fits are to the data points.